

Drawing causal relationships

Causation in biology

A key goal in science is understanding *how* the world works. Or put another way, we want to know what *causes* what. Understanding causation is at the heart of understanding how a system works (e.g., the physiology of an organism, or the growth of an epidemic in a population), but is also critical being able to intervene. If we were to change *this* thing (e.g., the level of this hormone), how would it change *that* thing down the line (e.g., blood pressure)? Indeed, that is a reasonable definition of causation: if we change x , does it change y ?

Drawing causal relationships

There are two difficulties in communicating causal relationships¹. First, words are so slippery! Statements like “regularly eating breakfast leads to increased happiness” leave a lot to the imagination. Does breakfast do this directly or through some other factor or mechanism, like energy level or blood sugar? Or is there some common cause of happiness and breakfast eating, like living in a home that has a regular schedule and enough time and money to make a good breakfast?

Second, in complex systems like those that we study in biology there are often *multiple* causes for any factor of interest. A person’s happiness² is likely affected by physical factors (e.g., are they in pain or uncomfortable), physiological factors (e.g., hormone status, blood sugar), psychological factors (e.g., confidence, sense of agency), social factors (e.g., sense of support,

¹ OK, at *least* two!

² Or health or any other metric

friend group), and so on. To say that one thing in this whole messy set of interactions “causes” happiness without acknowledging the rest is misleading at best.

Wouldn’t it be helpful if we had a way to sketch out the whole network of interacting factors that cause some variable of interest? We could use it to draw how we think the world works, like drawing a hypothesis. We could also use it to understand where a study fit in and what aspect(s) of the system it relates to. We could even compare our drawing with someone else to see if we were thinking about the system in the same way. In fact, there might be a whole lot to be learned from these drawings³!

Enter the DAG

There is a type of graph or diagram that is designed to do just: to show causal relationships. They are called “directed, acyclic graphs,” or DAGs for short. Those words are all sort of mathy jargon⁴. Don’t worry about the jargon. For our purpose, they are simply box-and-arrow graphs to describe *what causes what*⁵. They are beguilingly simple, but there are some rules we need to follow.

1. The boxes will represent the things that are playing a part in this causal story. They are things or variables or whatnot. In general they are quantities we could at least theoretically measure⁶
2. The arrows will represent *causal* relationships such that $x \rightarrow y$ implies that a change in x *causes* a change in y . Note that we are not drawing things in order, what comes first or second, but rather cause and effect relationships⁷.
3. The arrows do not imply a direction (positive or negative) or shape (linear, sigmoidal) to the causal relationship, just that there is one. Also, these arrows do *not* represent alternative paths! You would have one arrow that could imply an increase or a decrease in y depending on how the cause, x , is changed and the nature of the shape of the relationship between the two.⁸

³ For instance, they can help us understand which statistical models to use and what parameter estimates represent. You’ll see them in epidemiology, econometrics, sociology, and increasingly in ecological research. If you stay in science, you’ll see them. But whoops! I’m getting ahead of myself

⁴ Directed means having arrows, acyclic means no cycles, and graph means, to mathy folks, having nodes and edges (the arrows in this case).

⁵ We use graphs and diagrams to represent a *lot* of things in science: relationships between ideas (think mind maps), the flow of nutrients or individuals between states (think epidemic models, which we will see later), or the transformation of one molecule into another in, say, the Krebs cycle, to name a few. Just note that DAGs are one particular use of diagrams for a very specific purpose and try not to confuse it with those used for other purposes.

⁶ They are *not* processes, like grow or shrink or spread.

⁷ These *could* represent one or more processes, like accumulating or losing resources.

⁸ This is the rule that students most often get wrong.

4. We do not draw interactions. If this means nothing to you, fine, just move along to number 5. If it does, just note that diagrams with additive or interactive or contingent relationships all look the same.
5. No cycles⁹. That is, you should not be able to follow the arrows to get back to where you started from.

⁹ That’s the “acyclic” part of the the acronym.

Causation in my household

It helps to start with an example, even a quite stupid one. So here we go:

The other day I walked into the kitchen and found a puddle of water on the counter. Being a good scientist, in addition to an annoyed parent, I started to consider what caused this puddle to be there. One idea immediately leaped to mind: one of my kids had probably grabbed some ice for their drink and left a few pieces on the counter, which melted and formed a puddle.¹⁰ Now how to draw this? That is, after all, the reason for this story.

Here is what my initial graph might look like (Figure 1). In this case we would read this as, “my kid caused ice to be on the counter” and then “the ice caused water to be on the counter”.

But wait, there is probably at least a bit more. Ice doesn’t turn to water without melting, right? That requires heat! So we need to add “heat” to the diagram.

This second diagram (Figure 2) now adds, “a change in heat causes a change in water,” which sounds weird because this only makes sense when there is ice present, too. But our diagram is about the system (in my kitchen) as a whole. This arrow does not necessarily say that heat causes water *in general*. It just says that if we were to change the heat (e.g., remove it from the room) we would change the presence of liquid water on the counter. Note, however, that the graph does not *by itself* imply that you need both ice and heat to get water. It just sidesteps the whole issue. *We*, as thinking people, add that interpretation¹¹. So now we have arrows saying that if we were

¹⁰ Other examples in science might be more compelling, sure, but they always say to write what you know!

Kid → Ice → Water

Figure 1: A simple DAG.

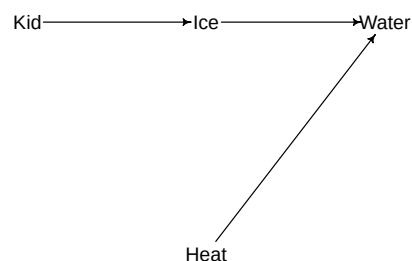


Figure 2: A less simple DAG.

¹¹ See rule 4, above, about “interactions.” This is what I meant.

to change the ice on the counter or the heat in the room we would cause a change in the water on the counter.

If your head hurts, that’s probably a good sign. It suggests you are paying attention. This is not always an initially intuitive way of describing a system or drawing a graph.

Anyway, we might need to complicate our diagram a bit, because I suspected that my kid wanted ice because it was hot out. In other words, it seemed like heat in the house on this warm day *caused* the ice to be on the counter, along with my sloppy kid. Thus this graph to the right.

What this graph (Figure 3) means is that a change in heat causes a change in water on my kitchen counter through two **causal paths**: 1) by causing the ice to be present or not, which then can turn into water ($Heat \rightarrow Ice \rightarrow Water$) and 2) by providing or not providing the energy to melt the ice ($Heat \rightarrow Water$). Having shapes like this is fine in these graphs just so long as the arrows do not form a cycle. We could even add an arrow for the causal effect of, say, my cat who likes to play with ice cubes. Where would you add this causal relationship?

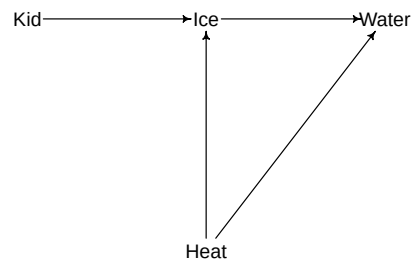


Figure 3: An even less simple DAG.

The point?

What, you may be asking, is the point of this? Well, setting aside all of the wonderfully useful logical and statistical magic this enables¹², which we won’t be using¹³, these diagrams force us to think very carefully about causation. They also help us think through “what if” scenarios, like “what if we added or removed the heat?” or “what if we added another kid?” That is, they help us see the consequences of **interventions**, which might include experimental manipulations or public health measures or anything else we might try. For instance, this diagram suggests that making sure there are ample paper towels would not change anything because we have no way for them to cause any changes¹⁴.

The other point to highlight is that this diagram is, in essence, a hypothesis. It both describes the hypothesis, the statement about how the world (or at least my kitchen) works, and what

¹² see the footnote, above

¹³ Phew!

¹⁴ But we could create a different DAG with toweling in it.

would change given particular interventions. But if this model is wrong, then so are its predictions about interventions.

Again, you might be asking, “well what’s the point” if it can be wrong. I have two responses. First, welcome to the real world where there are no magic wands¹⁵ that can tell us the Truth with a capital “T.” Second, as I will emphasize in this class, we simply cannot make sense of *anything* in the world without invoking some sort of model. We do this instinctively, telling stories, but drawing out a DAG forces you to be clearer and makes it easier for someone else (sometimes *you*) to see how you are thinking about the world. So yeah, the model can be wrong, but that’s not the model’s fault. *You* made the model! Also, stop worrying so much about being wrong. It’s unhealthy!

An alternative

There is absolutely nothing wrong with having an alternative working model in mind. In fact it is encouraged! So let’s consider an alternative explanation and DAG for the water on the counter: it’s rain water. In this scenario I’m thinking that the kid left the window open which let in the rain which then pooled on the counter. Like this one to the right (Figure 4).

Did that match your expectations? Oh, and we can add in heat, too, because heat would cause the water to evaporate, causing it to *not* be on the counter. Oh and maybe the kid opened the window because it was hot in the kitchen, so there might be an arrow there, too (Figure 5).

Before ending this silly example I would like to point out two related, perhaps not obvious consequence of this last DAG (Figure 5): 1) It suggests that if we knew the status of the window, whether it was open or closed, knowing whether the kid was in the kitchen opening or closing anything provides no extra information about whether we should see water on the counter. This will make more sense in larger studies with more data, but if you have data on the consequence (or **descendant**) of some **upstream** cause, then information on that upstream cause does not help you predict anything or describe anything in the data.

¹⁵ Or statistical or other tools.

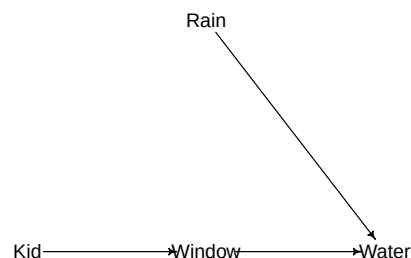


Figure 4: An alternate hypothesis

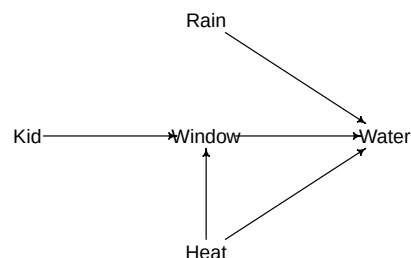


Figure 5: An alternate hypothesis, but now with heat!.

It's more of a statsy thing, but it may crop up in our work in this class. 2) If we were intervene by, say, nailing the window open or shut, we remove the influence of the kid on water on the counter, and of heat on the status of the window. Indeed, that's what experiments and interventions do: they remove dependencies in our system. In that case, our DAG would look like this last one on the right (Figure 6).

Let me summarize

So what we have now is a method to graph or diagram expected or assumed causal relationships between variables. These diagrams are helpful for clarifying our own thinking, but also for communicating our thinking to others. They are also helpful when trying to connect evidence to causal relationships, as in, *which arrow(s) did this study include or exclude or just ignore?* They are a tool I will ask you to use for several assignments and one that I would encourage you to use for others (e.g., the WPs). Like all tools, they take a bit of practice to get good with, so in addition to the specific opportunities I provide in this class, see if you can't use them elsewhere. You'll thank me, eventually!

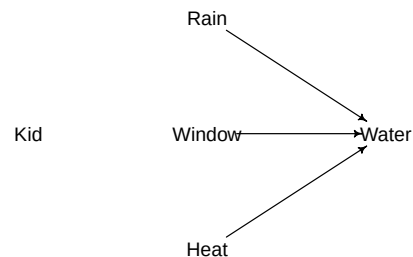


Figure 6: What if we nailed the window open?